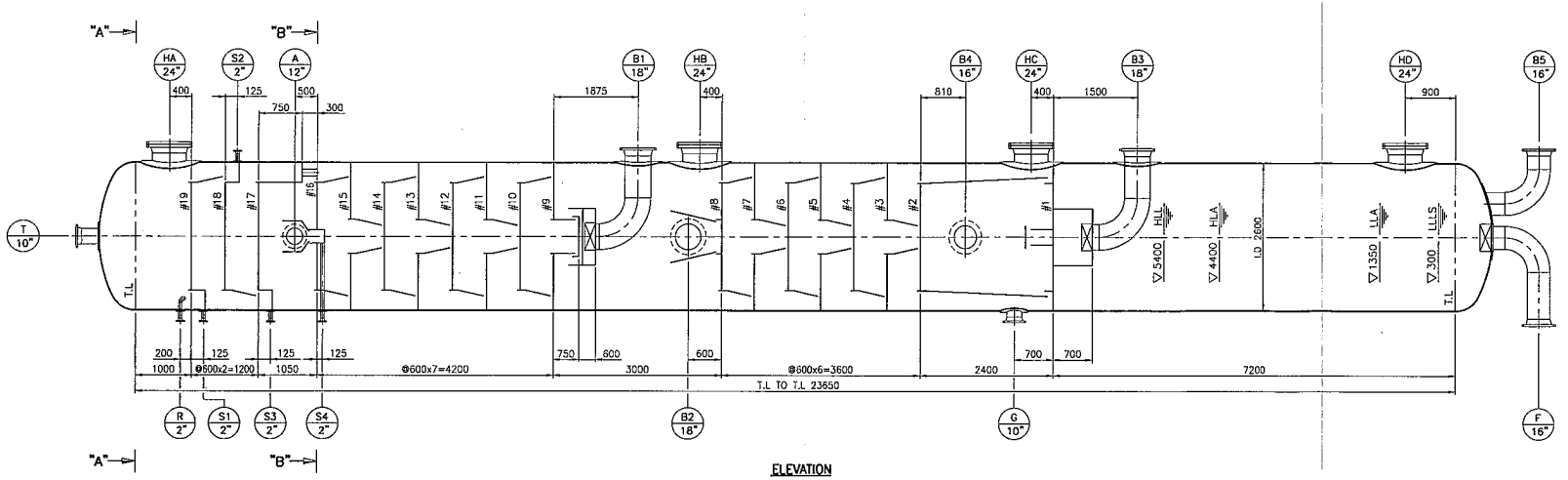
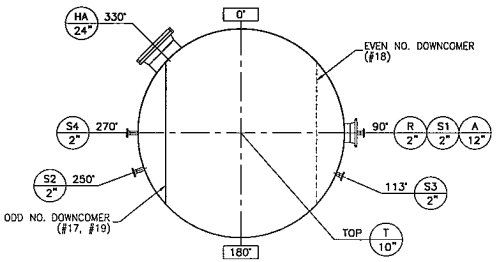


ELEVATION

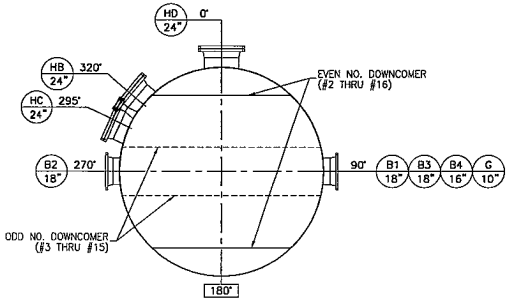
GENERAL INFORMATION

1. ALL DIMENSIONS ARE GIVEN IN MILLIMETERS.
2. ALL TOWER ATTACHMENTS SHOWN ON THIS DRAWING BE NOT ACTUAL FABRICATION DRAWING. THEREFORE, THE FABRICATION DRAWING SHOULD BE PREPARED BY VESSEL VENDOR.
3. ALL PARTS SHOWN ON THIS DRAWING SHALL BE SUPPLIED AND INSTALLED BY VESSEL FABRICATOR.
4. ALL TRAY ELEVATIONS ARE TO TOP OF THE TRAY SUPPORT RINGS.
5. SUPPORT RINGS ARE TO BE LEVEL ALL AROUND CIRCUMFERENCE OF TOWER, ACROSS TOWER, ACROSS SUPPORT RING FROM TOWER TO INNER EDGE.
6. GRIND OFF ALL BURRS AND SLUG ON THE TOP SURFACE OF SUPPORT RING.
7. REFERENCE DWG. NO. : VP-6340S-103-C1101-003 ~ 013

FINAL
2006. 6. 19
HANBAL MASSTECH LIMITED



ORIENTATION PLAN
SECTION "A-A"
(#17 THRU #19)



ORIENTATION PLAN
SECTION "B-B"
(#2 THRU #16)

REV.	DATE	DESCRIPTION	ORIG.	PREP.	CHKD.	APPD.	AUTH.
F	19/06/06	FINAL	HMT	Y.P.KIM	K.T.Moon	J.H.Kwon	
3	21/02/06	FOR FINAL	HMT	Y.P.KIM	K.T.Moon	J.H.Kwon	
2	28/12/05	FOR FINAL	HMT	Y.P.KIM	K.T.Moon	J.H.Kwon	
1	27/04/05	REVISED BY COMMENTS	HMT	Y.P.KIM	K.T.Moon	J.H.Kwon	
0	05/03/05	FOR APPROVAL	HMT	Y.P.KIM	K.T.Moon	J.H.Kwon	

SOUTH PARS GAS FIELD DEVELOPMENT-
PHASES 9 & 10, ASSALUYEH-IRAN
ONSHORE FACILITIES

N.I.O.C
PARS OIL AND GAS COMPANY

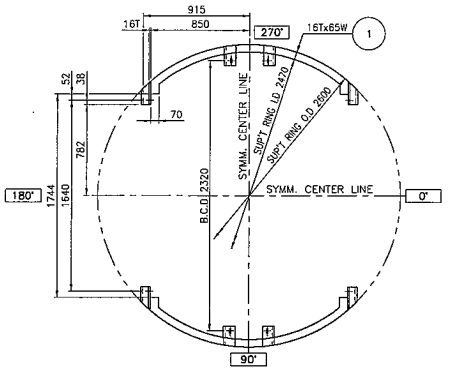
PROJECT No.: POGC-760-81-156 | DOC. CLASS. | SCALE : NONE

HMT | OIEC | IOEC

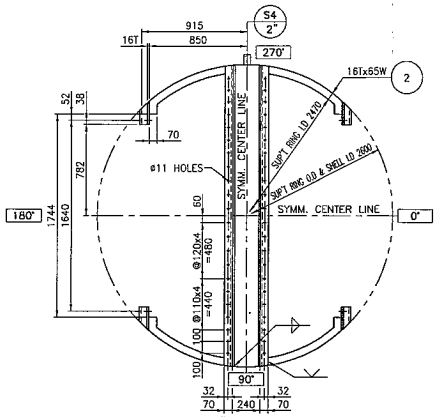
TITLE:
103-C-101/201
CONDENSATE STABILISER
VESSEL SKETCH

DWG No. VP-6340S-103-C1101-003 | SHEET No. 1 OF 11 | REV. F

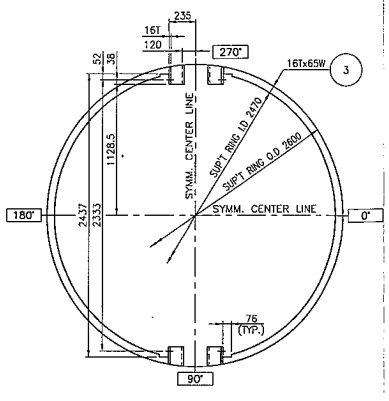
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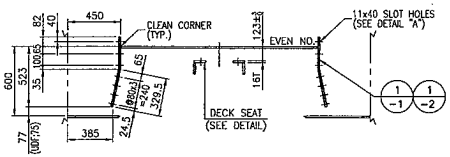
PLAN VIEW OF EVEN NO. TRAY SUPPORT RING(#2 THRU #14)



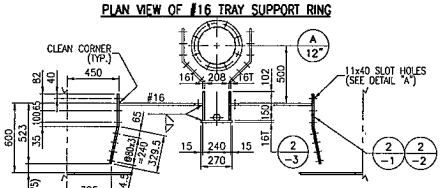
PLAN VIEW OF #16 TRAY SUPPORT RING



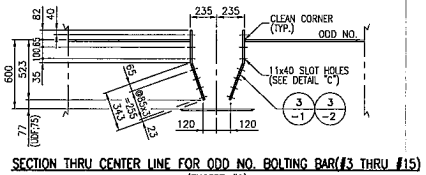
PLAN VIEW OF ODD NO. TRAY SUPPORT RING(#3 THRU #15)
(EXCEPT #9)



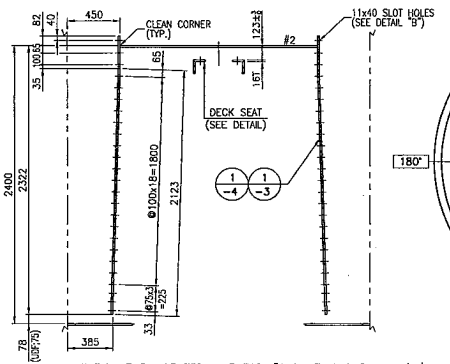
SECTION THRU CENTER LINE FOR EVEN NO. BOLTING BAR(#4 THRU #14)



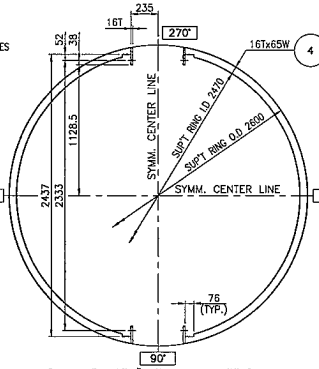
SECTION THRU CENTER LINE FOR #16 BOLTING BAR



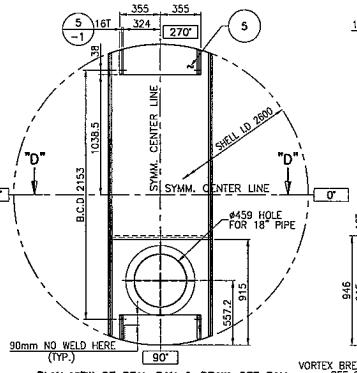
SECTION THRU CENTER LINE FOR ODD NO. BOLTING BAR(#3 THRU #15)
(EXCEPT #9)



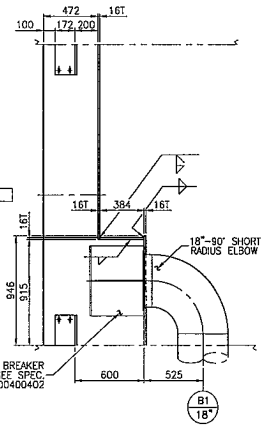
SECTION THRU CENTER LINE FOR #2 BOLTING BAR



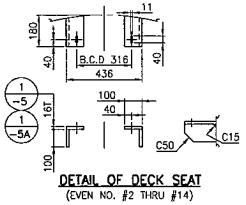
PLAN VIEW OF #9 TRAY SUPPORT RING



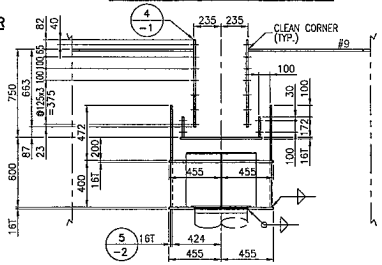
PLAN VIEW OF SEAL PAN & DRAW-OFF PAN



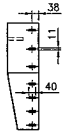
SECTION THRU CENTER LINE FOR SEAL PAN & DRAW-OFF PAN



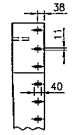
DETAIL OF DECK SEAT
(EVEN NO. #2 THRU #14)



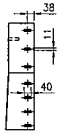
SECTION THRU CENTER LINE FOR #9 BOLTING BAR, SEAL PAN & DRAW-OFF PAN
VIEW "D-D"



DETAIL "A"



DETAIL "B"



DETAIL "C"

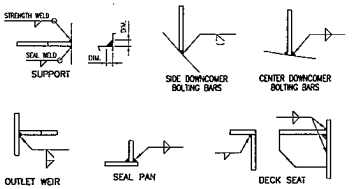
NO.		DESCRIPTION	MAT'L	Q'TY NET	Q'TY UNIT	REMARKS
1		SUPPORT RING	A516GR60N	7	SETS	16Tx65W
	-1	BOLTING BAR	A516GR60N	12	EA	16T
	-2	BOLTING BAR	A516GR60N	12	EA	16T
	-3	BOLTING BAR	A516GR60N	2	EA	16T
	-4	BOLTING BAR	A516GR60N	2	EA	16T
	-5	DECK SEAT	A516GR60N	14	EA	16T
	-5A	DECK SEAT	A516GR60N	14	EA	16T
2		SUPPORT RING	A516GR60N	1	SET	16Tx65W
	-1	BOLTING BAR	A516GR60N	2	EA	16T
	-2	BOLTING BAR	A516GR60N	2	EA	16T
	-3	SUMP	A516GR60N	1	EA	16T
3		SUPPORT RING	A516GR60N	6	SETS	16Tx65W
	-1	BOLTING BAR	A516GR60N	12	EA	16T
	-2	BOLTING BAR	A516GR60N	12	EA	16T
4		SUPPORT RING	A516GR60N	1	SET	16Tx65W
	-1	BOLTING BAR	A516GR60N	4	EA	16T
5		SEAL PAN SUPPORT	A516GR60N	2	SETS	16T
	-1	BOLTING BAR	A516GR60N	4	EA	16T
	-2	DRAW-OFF PAN	A516GR60N	1	SET	16T

(TOTAL; 2 SETS)

GENERAL INFORMATION

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4. ALL TRAY ELEVATIONS ARE TO TOP OF THE TRAY SUPPORT RINGS.
5. SUPPORT RINGS ARE TO BE LEVEL ALL AROUND CIRCUMFERENCE OF TOWER, ACROSS TOWER, ACROSS SUPPORT RING FROM TOWER TO INNER EDGE.
6. GRIND OFF ALL BURRS AND SLUG ON THE TOP SURFACE OF SUPPROT RING.
7. REFERENCE DWG. NO. : VP-6340S-103-C1101-003 ~ 013.

TYPICAL WELD DETAILS



NOTE : All welds are continuous 11mm fillet welds unless otherwise specified.

FINAL
2006. 6. 19
HANBAL MASSTECH LIMITED

REV.	DATE	DESCRIPTION	ORIG.	PREP.	CHKD.	APPD.	AUTH.
F	19/06/06	FINAL	HMT	Y.P.KIM	K.T.Moon	J.H.Kwon	
I	23/05/05	REVISED BY COMMENTS	HMT	Y.P.KIM	K.T.Moon	J.H.Kwon	
0	05/03/05	FOR APPROVAL	HMT	Y.P.KIM	K.T.Moon	J.H.Kwon	

SOUTH PARS GAS FIELD DEVELOPMENT-
PHASES 9 & 10, ASSALUYEH-IRAN
ONSHORE FACILITIES

N.I.O.C
PARS OIL AND GAS COMPANY

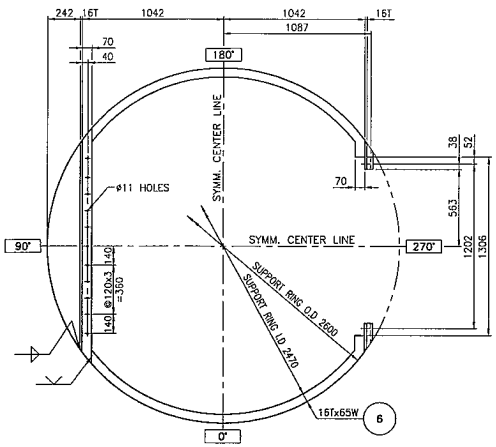
PROJECT No.: POGC-760-81-156	DOC. CLASS.	SCALE : NONE

HMT | OIEC | IOEC

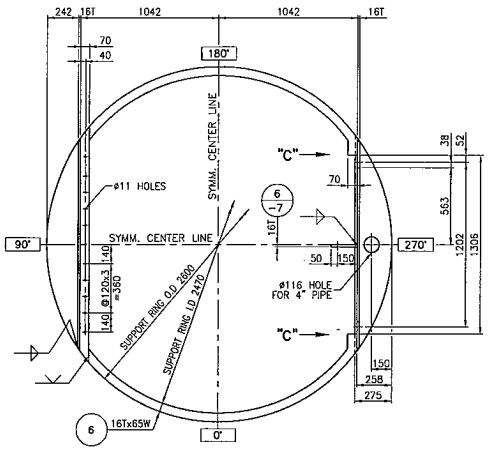
TITLE:
103-C-101/201
CONDENSATE STABILISER
TOWER ATTACHMENT DWG.(1/3)

DWG No.	VP-6340S-103-C1101-004	SHEET No. 2 OF 11	REV. F

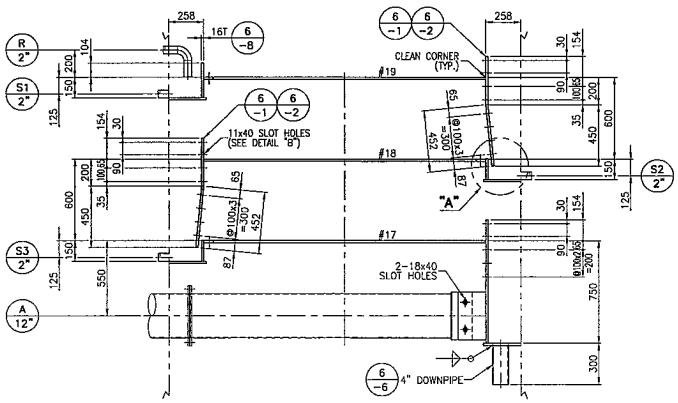
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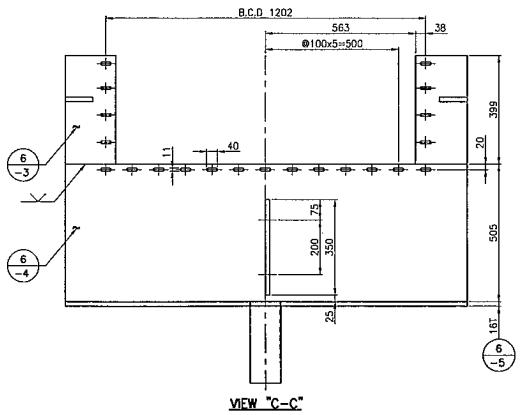
PLAN VIEW OF #19 TRAY SUPPORT RING
ROTATED 180° FOR #18 TRAY SUPPORT RING



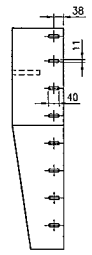
PLAN VIEW OF #17 TRAY SUPPORT RING



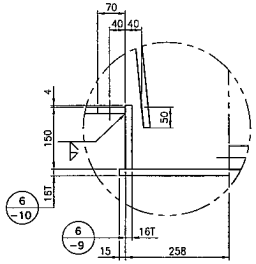
SECTION THRU CENTER LINE FOR #17, #18, #19 BOLTING BAR
& DOWNPIPE



VIEW "C-C"



DETAIL "B"



VIEW "A"

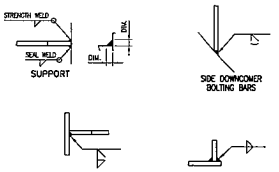
NO.		DESCRIPTION	MAT'L	Q'TY NO'T	Q'TY UNIT	REMARKS
6		SUPPORT RING	A516GR60N	3	SETS	16Tx65W
	-1	BOLTING BAR	A516GR60N	2	EA	16T
	-2	BOLTING BAR	A516GR60N	2	EA	16T
	-3	BOLTING BAR	A516GR60N	2	EA	16T
	-4	BOLTING BAR	A516GR60N	1	EA	16T
	-5	DECK	A516GR60N	1	EA	16T
	-6	DOWNPIPE	A312-TP304L	1	EA	4"
	-7	CLIP	A516GR60N	1	EA	16T
	-8	SIDE SUMP	A516GR60N	1	EA	16T
	-9	SIDE SUMP	A516GR60N	2	EA	16T
	-10	SIDE SUMP	A516GR60N	3	EA	16T

(TOTAL; 2 SETS)

GENERAL INFORMATION

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2. ALL TOWER ATTACHMENTS SHOWN ON THIS DRAWING BE NOT ACTUAL FABRICATION DRAWING. THEREFORE, THE FABRICATION DRAWING SHOULD BE PREPARED BY VESSEL VENDOR.
3. ALL PARTS SHOWN ON THIS DRAWING SHALL BE SUPPLIED AND INSTALLED BY VESSEL FABRICATOR.
4. ALL TRAY ELEVATIONS ARE TO TOP OF THE TRAY SUPPORT RINGS.
5. SUPPORT RINGS ARE TO BE LEVEL ALL AROUND CIRCUMFERENCE OF TOWER, ACROSS TOWER, ACROSS SUPPORT RING FROM TOWER TO INNER EDGE.
6. GRIND OFF ALL BURRS AND SLUG ON THE TOP SURFACE OF SUPPROT RING.
7. REFERENCE DWG. NO. : VP-6340S-103-C1101-003 ~ 013.

TYPICAL WELD DETAILS



NOTE : All welds are continuous 11mm fillet welds unless otherwise specified.

FINAL
2006. 6. 19
HANBAL MASSTECH LIMITED

REV.	DATE	DESCRIPTION	ORIG.	PREP.	CHKD.	APPD.	AUTH.
F	19/06/06	FINAL	HMT	Y.P.KIM	K.T.Moon	J.H.Kwon	
2	23/05/05	REVISED BY COMMENTS	HMT	Y.P.KIM	K.T.Moon	J.H.Kwon	
1	27/04/05	REVISED BY COMMENTS	HMT	Y.P.KIM	K.T.Moon	J.H.Kwon	
0	05/03/05	FOR APPROVAL	HMT	Y.P.KIM	K.T.Moon	J.H.Kwon	

SOUTH PARS GAS FIELD DEVELOPMENT-
PHASES 9 & 10, ASSALUYEH-IRAN
ONSHORE FACILITIES

N.I.O.C
PARS OIL AND GAS COMPANY

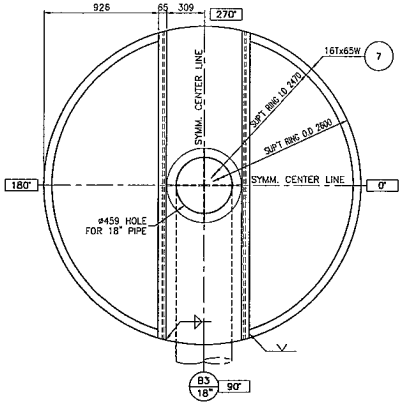
PROJECT No.: POGC-760-81-156	DOC. CLASS.	SCALE : NONE

HMT | OIEC | IOEC

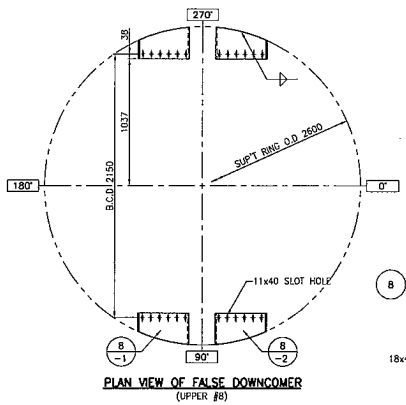
TITLE:
103-C-101/201
CONDENSATE STABILISER
TOWER ATTACHMENT DWG.(2/3)

DWG No.	VP-6340S-103-C1101-005	SHEET No. 3 OF 11	REV. F

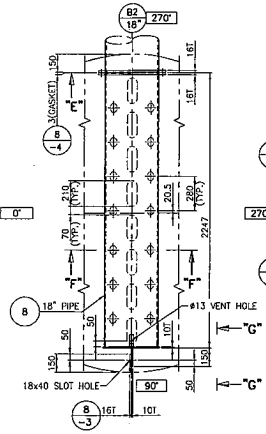
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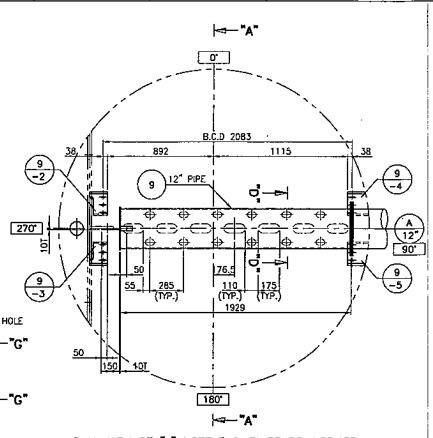
PLAN VIEW OF CHIMNEY TRAY SUPPORT RING
(BELOW #2)



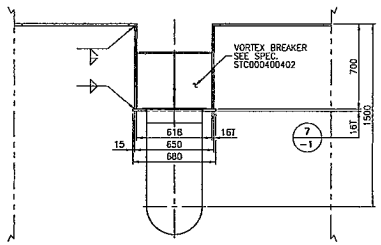
PLAN VIEW OF FALSE DOWNCOMER
(UPPER #8)



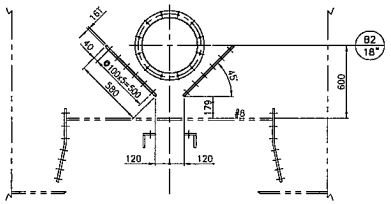
PLAN VIEW OF "B2" NOZZLE
(UPPER #8)



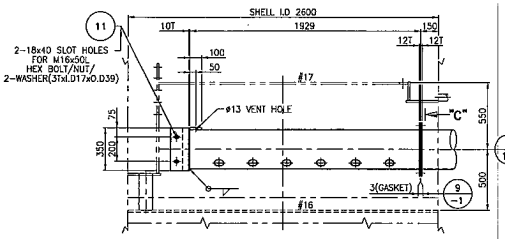
PLAN VIEW OF "A" NOZZLE & FALSE DOWNCOMER
(UPPER #16)



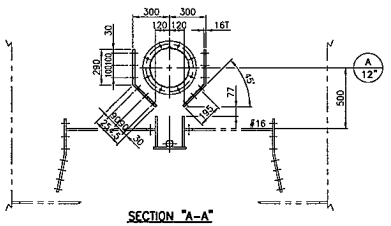
SECTION THRU CENTER LINE FOR CHIMNEY TRAY SUPPORT



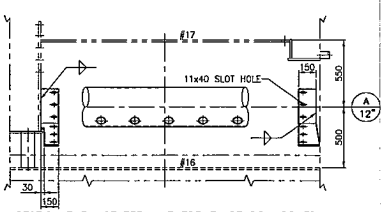
SECTION THRU CENTER LINE FOR FALSE DOWNCOMER



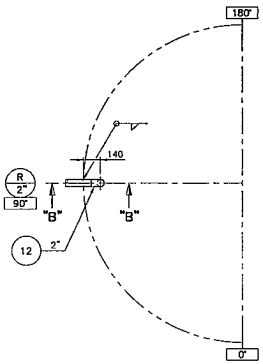
SECTION THRU CENTER LINE FOR "A" NOZZLE



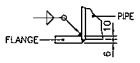
SECTION "A-A"



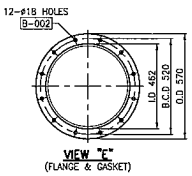
SECTION THRU CENTER LINE FOR FALSE DOWNCOMER



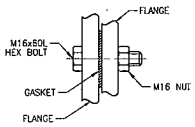
PLAN VIEW OF "R" NOZZLE
(UPPER #19)



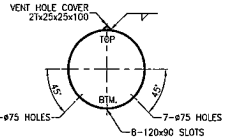
TYPICAL WELD DETAIL
(FOR "B2" NOZZLE)



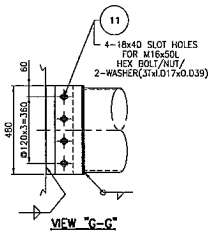
VIEW "C"
(FLANGE & GASKET)



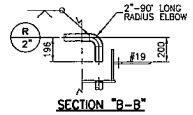
DETAIL "B-002" 13



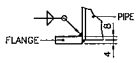
VIEW "F-F"



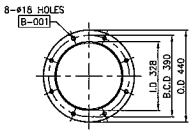
VIEW "G-G"



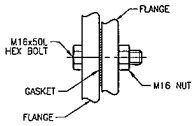
SECTION "B-B"



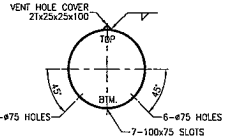
TYPICAL WELD DETAIL
(FOR "A" NOZZLE)



VIEW "C"
(FLANGE & GASKET)



DETAIL "B-001" 10



VIEW "D-D"

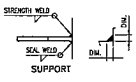
NO.		DESCRIPTION	MAT'L	Q'TY		REMARKS
				NET	UNIT	
7		SUPPORT RING	A516GR60N	1	SET	16Tx65W
	-1	SUMP	A516GR60N	1	EA	16T
8		FEED PIPE	A312-TP304L	1	SET	18" PIPE/16T/10T
	-1	FALSE DOWNCOMER	A516GR60N	2	EA	16T
	-2	FALSE DOWNCOMER	A516GR60N	2	EA	16T
	-3	CUP	A516GR60N	1	EA	16T
	-4	GSAKET	VALQUA-#6500	1	EA	3T(NON-ASBESTOS)
9		FEED PIPE	A312-TP304L	1	SET	12" PIPE/12T/10T
	-1	GSAKET	VALQUA-#6500	1	EA	3T(NON-ASBESTOS)
	-2	FALSE DOWNCOMER	A516GR60N	1	EA	16T
	-3	FALSE DOWNCOMER	A516GR60N	1	EA	16T
	-4	FALSE DOWNCOMER	A516GR60N	1	EA	16T
	-5	FALSE DOWNCOMER	A516GR60N	1	EA	16T
10		HEX BOLT/NUT	A193-B8/A194-8	8	SETS	M16x50L
11		HEX BOLT/NUT/2-WASHER	A193-B8/A194-8 A36-SS41	6	SETS	M16x50L/ 3T
12		FEED PIPE	A106-B	1	EA	12"
13		HEX BOLT/NUT	A193-B8/A194-8	12	SETS	M16x60L

(TOTAL; 2 SETS)

GENERAL INFORMATION

1. ALL DIMENSIONS ARE GIVEN IN MILLIMETERS.
2. ALL TOWER ATTACHMENTS SHOWN ON THIS DRAWING BE NOT ACTUAL FABRICATION DRAWING. THEREFORE, THE FABRICATION DRAWING SHOULD BE PREPARED BY VESSEL VENDOR.
3. ALL PARTS SHOWN ON THIS DRAWING SHALL BE SUPPLIED AND INSTALLED BY VESSEL FABRICATOR.
4. ALL TRAY ELEVATIONS ARE TO TOP OF THE TRAY SUPPORT RINGS.
5. SUPPORT RINGS ARE TO BE LEVEL ALL AROUND CIRCUMFERENCE OF TOWER, ACROSS TOWER, ACROSS SUPPORT RING FROM TOWER TO INNER EDGE.
6. GRIND OFF ALL BURRS AND SLUG ON THE TOP SURFACE OF SUPPROT RING.
7. REFERENCE DWG. NO. : VP-6340S-103-C1101-003 ~ 013.

TYPICAL WELD DETAILS



NOTE : All welds are continuous 11mm fillet welds unless otherwise specified.

FINAL
2006. 6. 19
HANBAL MASSTECH LIMITED

REV.	DATE	DESCRIPTION	ORIG.	PREP.	CHKD.	APPD.	AUTH.
F	19/06/06	FINAL	HMT	Y.P.KIM	K.T.Moon	J.H.Kwon	
2	23/06/05	REVISED BY COMMENTS	HMT	Y.P.KIM	K.T.Moon	J.H.Kwon	
1	27/04/05	REVISED BY COMMENTS	HMT	Y.P.KIM	K.T.Moon	J.H.Kwon	
0	05/03/05	FOR APPROVAL	HMT	Y.P.KIM	K.T.Moon	J.H.Kwon	

SOUTH PARS GAS FIELD DEVELOPMENT-
PHASES 9 & 10, ASSALUYEH-IRAN
ONSHORE FACILITIES

N.I.O.C
PARS OIL AND GAS COMPANY

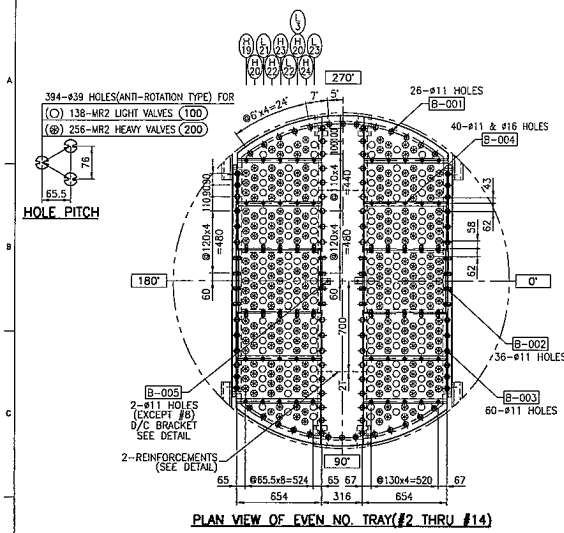
PROJECT No.: POGC-760-81-156	DOC. CLASS.	SCALE : NONE

HMT | OIEC | IOEC

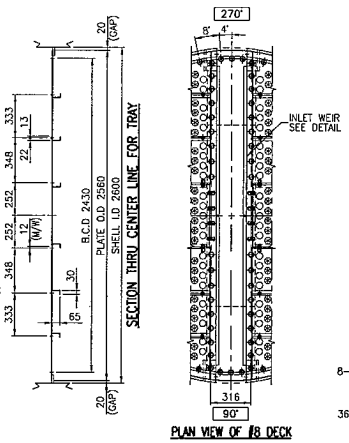
TITLE:
103-C-101/201
CONDENSATE STABILISER
TOWER ATTACHMENT DWG.(3/3)

DWG No.	VP-6340S-103-C1101-005	SHEET No. 4 OF 11	REV. F

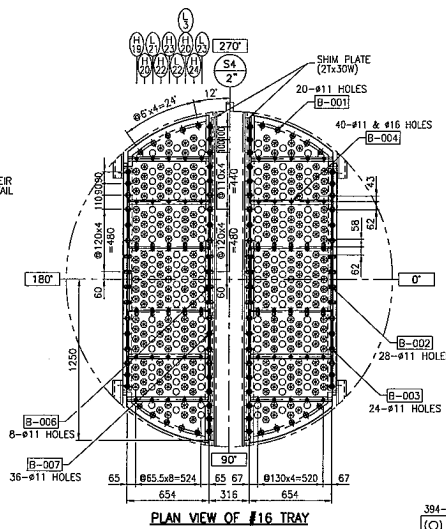
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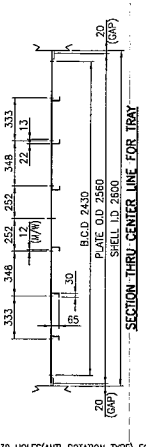
PLAN VIEW OF EVEN NO. TRAY(#2 THRU #14)



PLAN VIEW OF #8 DECK



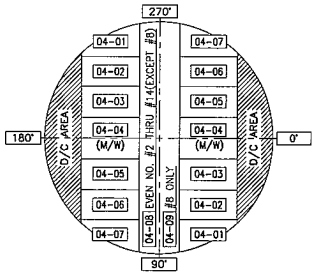
PLAN VIEW OF #16 TRAY



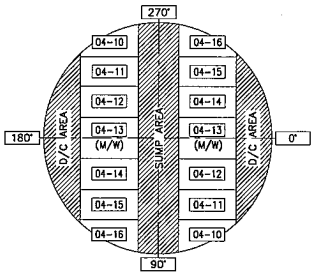
NO.	DESCRIPTION	MAT'L	QTY NET	QTY UNIT	REMARKS
					VP-6340S-103-C1101-012, 013.

GENERAL INFORMATION

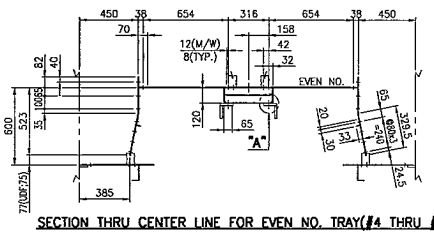
1. ALL DIMENSIONS ARE GIVEN IN MILLIMETERS.
2. TRAY PARTS ARE DESIGNED TO PASS THROUGH 24"(575mm) MANHOLE.
3. REFERENCE DWG. NO. : VP-6340S-103-C1101-003 ~ 013.



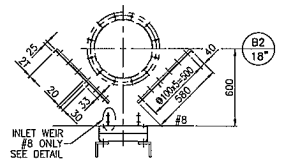
KEY PLAN EVEN NO. TRAY(#2 THRU #14)



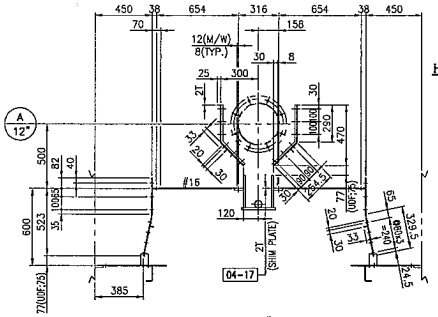
KEY PLAN #16 TRAY



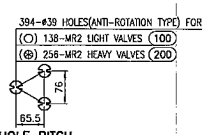
SECTION THRU CENTER LINE FOR EVEN NO. TRAY(#4 THRU #14)

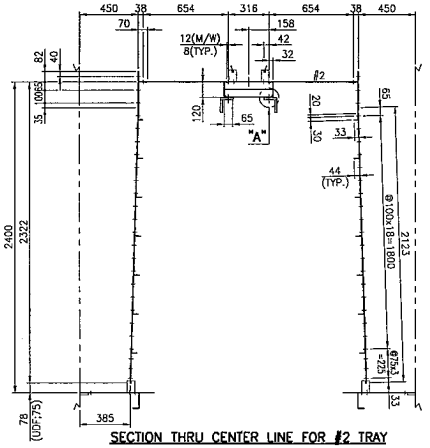


SECTION THRU CENTER LINE FOR FALSE DOWNCOMER

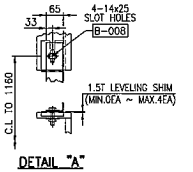


SECTION THRU CENTER LINE FOR #16 TRAY & FALSE DOWNCOMER

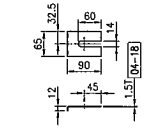




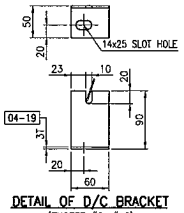
SECTION THRU CENTER LINE FOR #2 TRAY



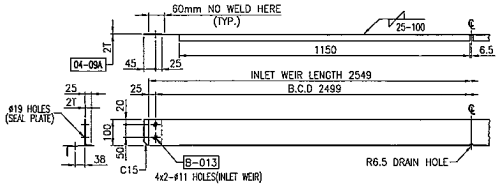
DETAIL "A"



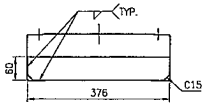
DETAIL OF LEVELING SHIM
(FOR EVEN NO. #2 THRU #14)



DETAIL OF D/C BRACKET
(EXCEPT #8, #16)



DETAIL OF INLET WEIR
(#8 ONLY)



DETAIL OF REINFORCEMENTS

FINAL
2006. 6. 19
HANBAL MASSTECH LIMITED

REV.	DATE	DESCRIPTION	ORIG.	PREP.	CHKD.	APPD.	AUTH.
F	19/06/06	FINAL	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	
1	23/07/05	REVISED BY COMMENTS	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	
0	13/06/05	FOR APPROVAL	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	

SOUTH PARS GAS FIELD DEVELOPMENT-
PHASES 9 & 10, ASSALUYEH-IRAN
ONSHORE FACILITIES

N.I.O.C
PARS OIL AND GAS COMPANY

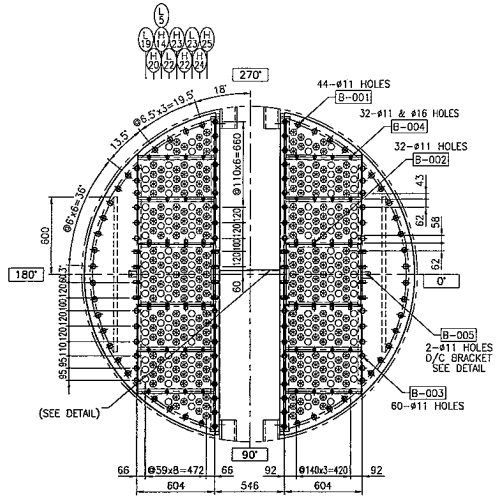
PROJECT No.: POGC-760-81-156	DOC. CLASS.	SCALE : NONE

HMT | OIEC | IOEC

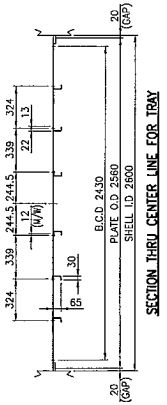
TITLE:
103-C-101/201
CONDENSATE STABILISER
TRAY DETAIL DRAWING (1/5)

DWG No.	VP-6340S-103-C1101-007	SHEET No. 5 OF 11	REV. F

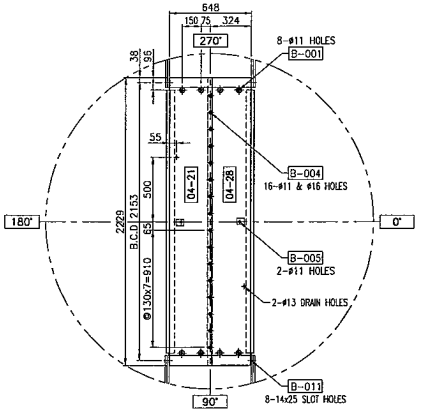
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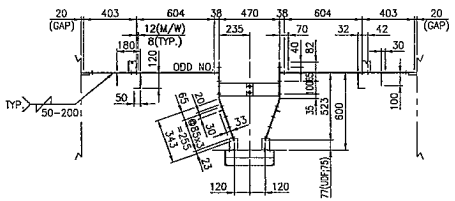
PLAN VIEW OF ODD NO. TRAY(#3 THRU #15)



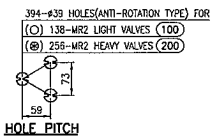
SECTION THRU CENTER LINE FOR TRAY



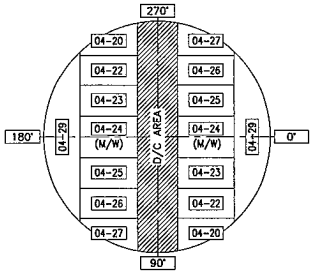
PLAN VIEW OF SEAL PAN
(BELOW #9)



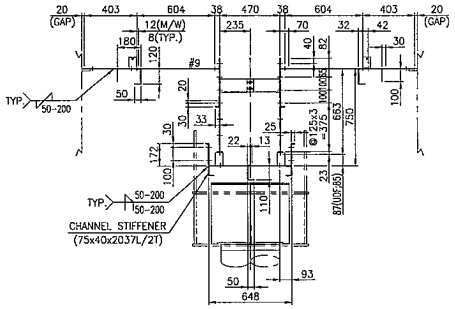
SECTION THRU CENTER LINE FOR ODD NO. TRAY(#3 THRU #15)
(EXCEPT #9)



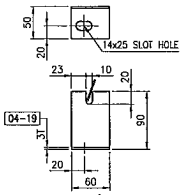
HOLE PITCH



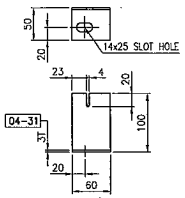
KEY PLAN ODD NO. TRAY(#3 THRU #15)



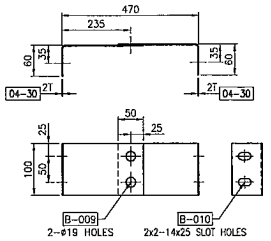
SECTION THRU CENTER LINE FOR #9 TRAY
& SEAL PAN



DETAIL OF D/C BRACKET



DETAIL OF D/C BRACKET
(FOR SEAL PAN)



DETAIL OF DOWNCOMER BRACES

GENERAL INFORMATION

1. ALL DIMENSIONS ARE GIVEN IN MILLIMETERS.
2. TRAY PARTS ARE DESIGNED TO PASS THROUGH 24"(575mm) MANHOLE.
3. REFERENCE DWG. NO. : VP-6340S-103-C101-003 ~ 013.

NO.	DESCRIPTION	MAT'L	Q'TY NET	Q'TY UNIT	REMARKS
					VP-6340S-103-C101-012, 013.

FINAL
2006. 6. 19
HANBAL MASSTECH LIMITED

REV.	DATE	DESCRIPTION	ORIG.	PREP.	CHKD.	APPD.	AUTH.
F	19/06/06	FINAL	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	
1	23/07/05	REVISED BY COMMENTS	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	
0	13/06/05	FOR APPROVAL	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	

SOUTH PARS GAS FIELD DEVELOPMENT-
PHASES 9 & 10, ASSALUYEH-IRAN
ONSHORE FACILITIES

N.I.O.C
PARS OIL AND GAS COMPANY

PROJECT No.: POGC-760-81-156 | DOC. CLASS. | SCALE : NONE

HMT | OIEC | IOEC

TITLE:
103-C-101/201
CONDENSATE STABILISER
TRAY DETAIL DRAWING (2/5)

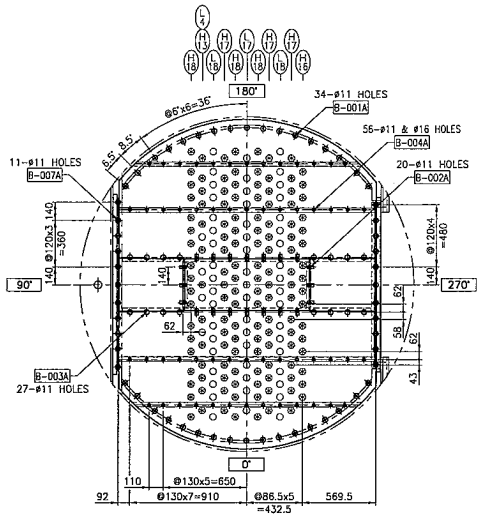
DWG No. VP-6340S-103-C101-008 | SHEET No. 6 OF 11 | REV. F

THIS DRAWING CONTAINS CONFIDENTIAL AND PROPRIETARY INFORMATION AND SHALL NOT BE COPIED OR OTHERWISE DIVULGED IN WHOLE OR IN PART WITHOUT THE EXPRESS WRITTEN CONSENT OF HANBAL MASSTECH LIMITED.

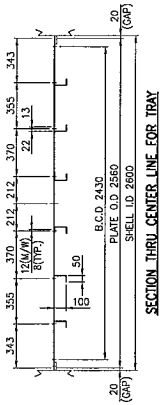
NO.	DESCRIPTION	MAT'L	Q'TY NET	Q'TY UNIT	REMARKS
					VP-6340S-103-C1101-012, 013

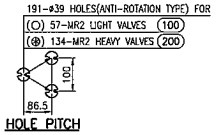
GENERAL INFORMATION

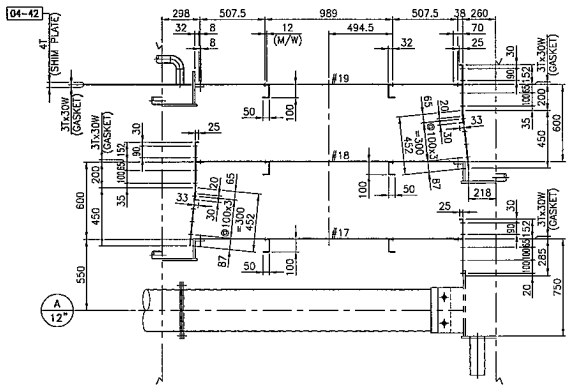
1. ALL DIMENSIONS ARE GIVEN IN MILLIMETERS.
2. TRAY PARTS ARE DESIGNED TO PASS THROUGH 24"(575mm) MANHOLE.
3. REFERENCE DWG. NO. : VP-6340S-103-C1101-003 ~ 013.



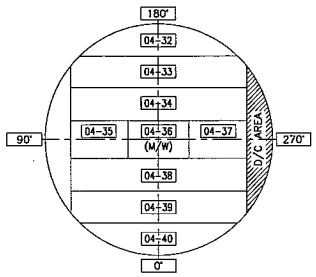
PLAN VIEW OF #17, #19 TRAY
ROTATED 180° FOR #18 TRAY







SECTION THRU CENTER LINE FOR #17, #18, #19 TRAY



KEY PLAN #17, #19 TRAY
ROTATED 180° FOR #18 TRAY

FINAL
2006. 6. 19
HANBAL MASSTECH LIMITED

REV.	DATE	DESCRIPTION	ORIG.	PREP.	CHKD.	APPD.	AUTH.
F	19/06/06	FINAL	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	
1	23/07/05	REVISED BY COMMENTS	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	
0	13/06/05	FOR APPROVAL	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	

SOUTH PARS GAS FIELD DEVELOPMENT-
PHASES 9 & 10, ASSALUYEH-IRAN
ONSHORE FACILITIES

N I O C
PARS OIL AND GAS COMPANY

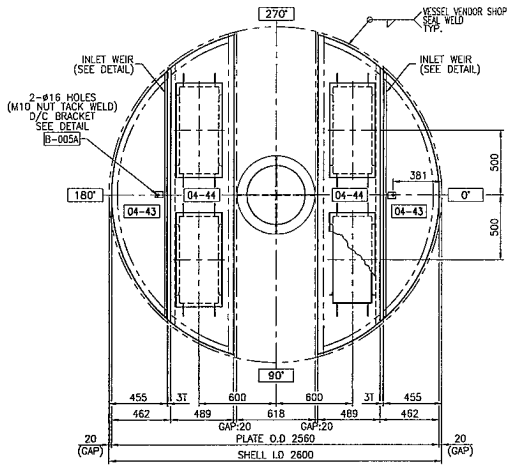
PROJECT No.: POGC-760-81-156 | DOC. CLASS. | SCALE : NONE

HMT | OIEC | I O E C

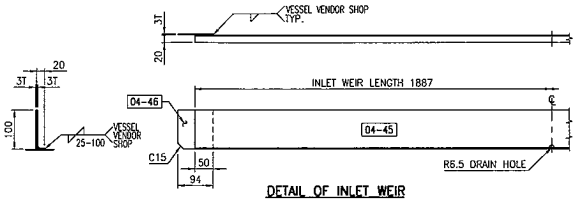
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TITLE:
103-C-101/201
CONDENSATE STABILISER
TRAY DETAIL DRAWING (3/5)

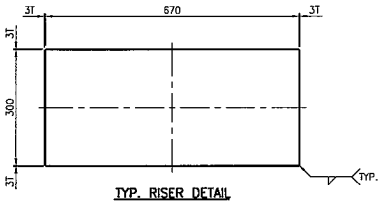
DWG No.	SHEET	REV.
VP-6340S-103-C1101-009	SHEET No. 7 OF 11	F



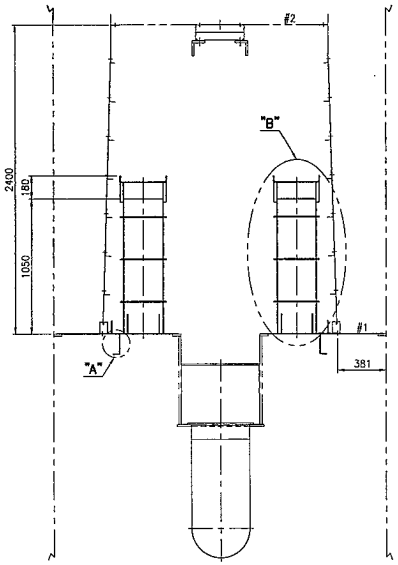
PLAN VIEW OF CHIMNEY TRAY
(BELOW #2)



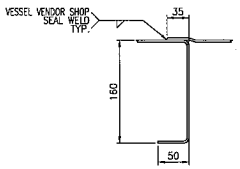
DETAIL OF INLET WEIR



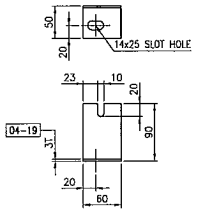
TYP. RISER DETAIL



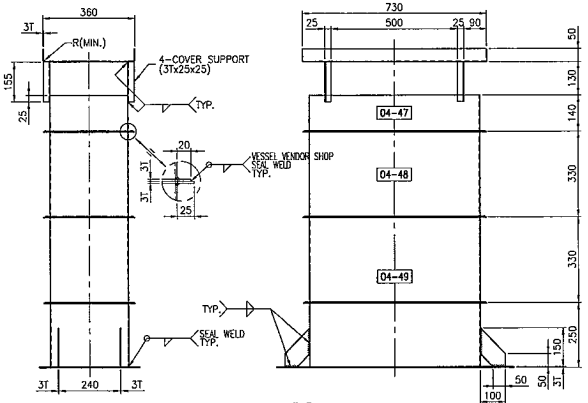
SECTION THRU CENTER LINE FOR CHIMNEY TRAY



DETAIL "A"



DETAIL OF D/C BRACKET



DETAIL "B"

NO.	DESCRIPTION	MAT'L	Q'TY NET	Q'TY UNIT	REMARKS
					VP-6340S-103-C1101-012, 013

GENERAL INFORMATION

1. ALL DIMENSIONS ARE GIVEN IN MILLIMETERS.
2. TRAY PARTS ARE DESIGNED TO PASS THROUGH 24"(575mm) MANHOLE.
3. REFERENCE DWG. NO. : VP-6340S-103-C1101-003 ~ 013.
4. CHIMNEY TRAY ARE FURNISHED BY HMT. INSTALLED BY VESSEL VENDOR.

FINAL
2006. 6. 19
HANBAL MASSTECH LIMITED

REV.	DATE	DESCRIPTION	ORIG.	PREP.	CHKD.	APPD.	AUTH.
F	19/06/06	FINAL	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	
1	23/07/05	REVISED BY COMMENTS	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	
0	13/06/05	FOR APPROVAL	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	

SOUTH PARS GAS FIELD DEVELOPMENT-
PHASES 9 & 10, ASSALUYEH-IRAN
ONSHORE FACILITIES

N.I.O.C
PARS OIL AND GAS COMPANY

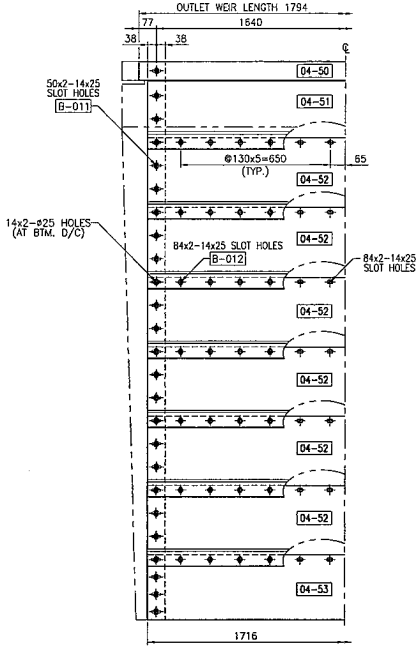
PROJECT No.: POGC-760-81-156 | DOC. CLASS. | SCALE : NONE

HMT | OIEC | IOEC

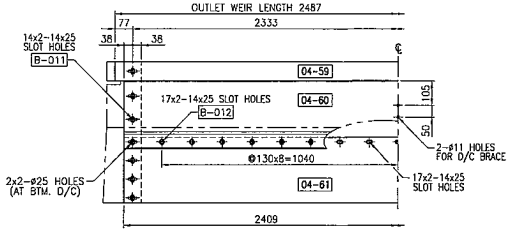
TITLE: 103-C-101/201
CONDENSATE STABILISER
TRAY DETAIL DRAWING (4/5)

DWG No. VP-6340S-103-C1101-010 | SHEET No. 8 OF 11 | REV. F

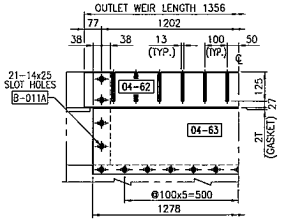
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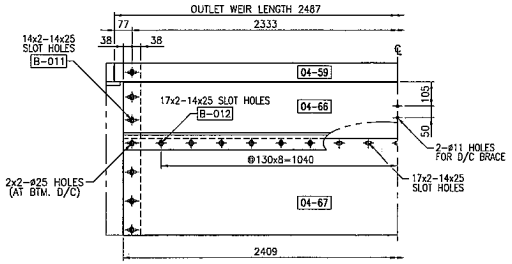
DETAIL OF #2 SIDE DOWNCOMER



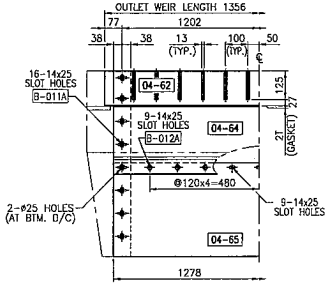
DETAIL OF CENTER DOWNCOMER(#3 THRU #15)
(EXCEPT #9)



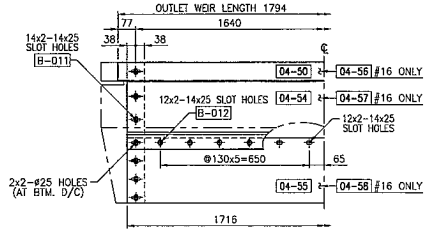
DETAIL OF #17 DOWNCOMER



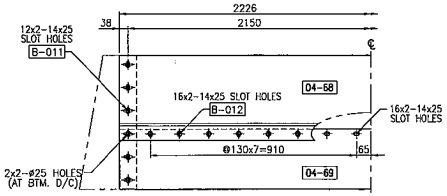
DETAIL OF CENTER DOWNCOMER(#9 ONLY)



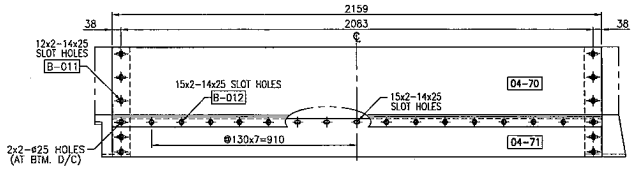
DETAIL OF #18, #19 DOWNCOMER



DETAIL OF SIDE DOWNCOMER(#4 THRU #16)



DETAIL OF FALSE DOWNCOMER
(ABOVE #8)



DETAIL OF FALSE DOWNCOMER
(ABOVE #16)

NO.	DESCRIPTION	MAT'L	Q'TY NET	Q'TY UNIT	REMARKS
					VP-6340S-103-C1101-012, 013.

GENERAL INFORMATION

1. ALL DIMENSIONS ARE GIVEN IN MILLIMETERS.
2. TRAY PARTS ARE DESIGNED TO PASS THROUGH 24"(575mm) MANHOLE.
3. REFERENCE DWG. NO. : VP-6340S-103-C1101-003 ~ 013.

FINAL
2006. 6. 19
HANBAL MASSTECH LIMITED

REV.	DATE	DESCRIPTION	ORIG.	PREP.	CHKD.	APPD.	AUTH.
F	19/06/06	FINAL	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	
1	23/07/05	REVISED BY COMMENTS	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	
0	13/06/05	FOR APPROVAL	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	

SOUTH PARS GAS FIELD DEVELOPMENT-
PHASES 9 & 10, ASSALUYEH-IRAN
ONSHORE FACILITIES

N.I.O.C
PARS OIL AND GAS COMPANY

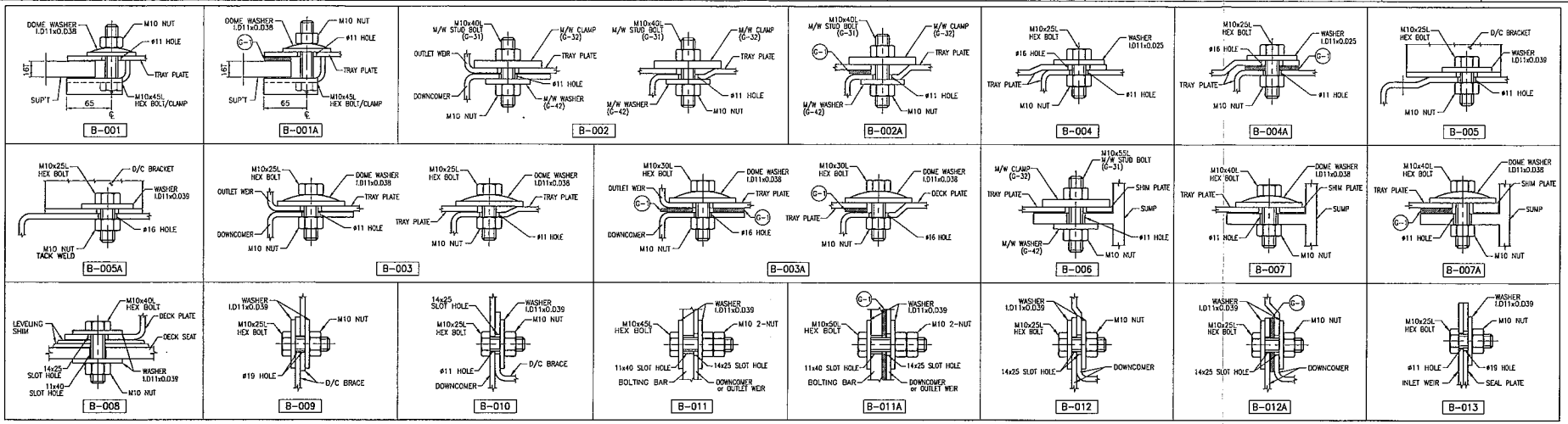
PROJECT No.: POGC-760-81-156 | DOC. CLASS. | SCALE : NONE

HMT | OIEC | IOEC

TITLE:
103-C-101/201
CONDENSATE STABILISER
TRAY DETAIL DRAWING (5/5)

DWG No. VP-6340S-103-C1101-011 | SHEET No. 9 OF 11 | REV. F

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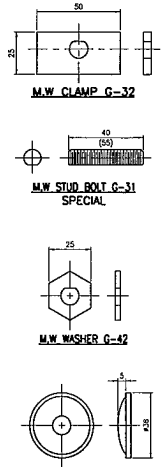


PARTS LIST

PART NO.	DESCRIPTION	MATERIAL	SIZE	NET Q'TY	SPARES (CON. & COMM.)	SPARES (2 YEARS)	TOTAL Q'TY	Q'TY PER INTERNAL: EVEN NO. TRAY (#2 ONLY) (1)	EVEN NO. TRAY (#4 THRU #14) (EXCEPT #6) (5)	EVEN NO. TRAY (#6 ONLY) (1)	EVEN NO. TRAY (#16 ONLY) (1)	ODD NO. TRAY (#3 THRU #15) (EXCEPT #9) (6)	ODD NO. TRAY (#9 ONLY) (1)	SEAL PAN (BELOW #9) (1)	TRAY (#17 ONLY) (1)	TRAY (#18, #19) (2)	CHIMNEY TRAY (BELOW #2) (1)
G-1	GASKET	FIBER-GLASS	31x30W	73.8M	18.5M	-	92.3M								24.4M	24.7M	
G-31	M/W STUD BOLT	A193-B8	M10x40L	476	48	-	524	36	36	36		32	32				
G-31	M/W STUD BOLT	A193-B8	M10x55L	8	10	-	18						8				
(B-001)	HEX BOLT/CLAMP	A193-B8/A240-304L	M10x45L/2T	498	50	-	548	26	26	26		44	44	8			
	HEX BOLT	A193-B8	M10x45L	496	50	-	546	100	28	52		28	28	8			
	HEX BOLT	A193-B8	M10x40L	64	10	-	74	4	4	4			36				
	HEX BOLT	A193-B8	M10x25L	2022	203	-	2225	270	126	164		134	134	18			2
	HEX NUT	A194-8	M10 NUT	4542	455	-	4997	572	284	370		298	350	42			
	HEX NUT(TACK WELD)	A194-8	M10 NUT	2	-	-	2										2
G-32	M/W CLAMP	A240-304L	5T	484	49	-	533	36	36	36		32	40				
G-42	M/W WASHER	A240-304L	2T	484	49	-	533	36	36	36		32	40				
K-74452	DOME WASHER	A240-304L	2Tx(.D11x0.D38	1374	138	-	1512	86	86	86		104	140	8			
	FLAT WASHER	A240-304L	2Tx(.D11x0.D39	2306	231	-	2537	546	114	232		134	134	18			2
	FLAT WASHER	A240-304L	2Tx(.D11x0.D25	520	52	-	572	40	40	40		32	32	16			
100	LIGHT VALVE	A240-304L	MR2	1932	97	-	2029	138	138	138		138	138				
200	HEAVY VALVE	A240-304L	MR2	3584	180	-	3764	256	256	256		256	256				
G-31	M/W STUD BOLT	A193-B8T	M10x40L	88	10	-	98				28				20	20	
G-31	M/W STUD BOLT	A193-B8T	M10x55L	8	10	-	18				8						
(B-001)	HEX BOLT/CLAMP	A193-B8T/A240-321	M10x45L/2T	20	5	-	25				20						
(B-001A)	HEX BOLT/CLAMP	A193-B8T/A240-321	M10x45L/2T	102	11	-	113								34	34	
	HEX BOLT	A193-B8T	M10x50L	53	10	-	63								21	16	
	HEX BOLT	A193-B8T	M10x45L	52	10	-	62				52						
	HEX BOLT	A193-B8T	M10x40L	69	10	-	79				36				11	11	
	HEX BOLT	A193-B8T	M10x30L	81	10	-	91								27	27	
	HEX BOLT	A193-B8T	M10x25L	304	31	-	335				118				56	65	
	HEX NUT	A194-8T	M10 NUT	978	98	-	1076				350				210	209	
G-32	M/W CLAMP	A240-321	5T	96	10	-	106				36				20	20	
G-42	M/W WASHER	A240-321	2T	96	10	-	106				36				20	20	
K-74452	DOME WASHER	A240-321	2Tx(.D11x0.D38	296	30	-	326				80				72	72	
	FLAT WASHER	A240-321	2Tx(.D11x0.D39	354	36	-	390				212				42	50	
	FLAT WASHER	A240-321	2Tx(.D11x0.D25	208	21	-	229				40				56	56	
100	LIGHT VALVE	A240-321	MR2	309	16	-	325				138				57	57	
200	HEAVY VALVE	A240-321	MR2	658	33	-	691				256				134	134	

ASSEMBLIES PER INTERNAL

ASSEMBLY NUMBER	EVEN NO. TRAY (#2 ONLY) (1)	EVEN NO. TRAY (#4 THRU #14) (EXCEPT #6) (5)	EVEN NO. TRAY (#6 ONLY) (1)	EVEN NO. TRAY (#16 ONLY) (1)	ODD NO. TRAY (#3 THRU #15) (EXCEPT #9) (6)	ODD NO. TRAY (#9 ONLY) (1)	SEAL PAN (BELOW #9) (1)	TRAY (#17 ONLY) (1)	TRAY (#18, #19) (2)	CHIMNEY TRAY (BELOW #2) (1)	TOTAL
B-001	26	26	26	20	44	44	8				518
B-001A								34	34		102
B-002	36	36	36	28	32	32					504
B-002A								20	20		60
B-003	60	60	60	24	60	60					864
B-003A								27	27		81
B-004	40	40	40	40	32	32	16				560
B-004A								56	56		168
B-005	2	2			2	2	2				28
B-005A										2	2
B-006				8		8					16
B-007				36		36					72
B-007A								11	11		33
B-008	4	4	4								28
B-009					2	2					14
B-010					4	4					28
B-011	100	28	52	52	28	28	8				548
B-011A								21	16		53
B-012	168	24	56	54	34	34					636
B-012A									9		18
B-013			8								8



GENERAL INFORMATION

1. ALL DIMENSIONS ARE GIVEN IN MILLIMETERS.
2. REFERENCE DWG. NO. : VP-6340S-103-C1101-003 ~ 013.
3. SPARE PARTS
   1) FOR COMMISSIONING / CONSTRUCTION
   - BOLT/NUT/WASHER : 10% (MIN. 10 PCS OF EACH TYPE)
   - CLAMP : 10% (MIN. 5 PCS)
   - GASKET : 25% (10 FEET)
   - VALVE CAPS : 5%

FINAL
2006. 6. 19
HANBAL MASSTECH LIMITED

REV.	DATE	DESCRIPTION	ORIG.	PREP.	CHKD.	APPD.	AUTH.
F	19/05/06	FINAL	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	
1	23/07/05	REVISED BY COMMENTS	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	
0	13/06/05	FOR APPROVAL	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	

SOUTH PARS GAS FIELD DEVELOPMENT-
PHASES 9 & 10, ASSALUYEH-IRAN
ONSHORE FACILITIES

N.I.O.C
PARS OIL AND GAS COMPANY

PROJECT No.: PGGC-760-81-156 | DOC. CLASS. | SCALE : NONE

HMT | OIEC | IOEC

TITLE: 103-C-101/201
CONDENSATE STABILISER
BOLTING ASS'Y DWG.

DWG No. VP-6340S-103-C1101-012 | SHEET No. 10 OF 11 | REV. F

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PARTS LIST

PART NO.	DESCRIPTION	MATERIAL	SIZE	NET Q'TY	SPARES (CON. & COMM.)	SPARES (2 YEARS)	TOTAL Q'TY	Q'TY PER INTERNAL: EVEN NO. TRAY (#2 THRU #16)	EVEN NO. TRAY (#4 THRU #14)	EXCEPT #8	ODD NO. TRAY (#8 ONLY)	EVEN NO. TRAY (#16 ONLY)	ODD NO. TRAY (#3 THRU #15) EXCEPT #9	ODD NO. TRAY (#9 ONLY)	SEAL PAN (BELOW #9)	TRAY (#17 ONLY)	TRAY (#18, #19)	CHIMNEY TRAY (BELOW #2)
								(1)	(5)	(1)	(1)	(1)	(6)	(1)	(1)	(1)	(2)	(1)
04-01	TRAY PLATE	A240-304L	2T	14	–	–	14	2	2	2								
04-02	TRAY PLATE	A240-304L	2T	14	–	–	14	2	2	2								
04-03	TRAY PLATE	A240-304L	2T	14	–	–	14	2	2	2								
04-04	TRAY PLATE(M/W)	A240-304L	2T	14	–	–	14	2	2	2								
04-05	TRAY PLATE	A240-304L	2T	14	–	–	14	2	2	2								
04-06	TRAY PLATE	A240-304L	2T	14	–	–	14	2	2	2								
04-07	TRAY PLATE	A240-304L	2T	14	–	–	14	2	2	2								
04-08	CENTER DECK	A240-304L	2T	6	–	–	6	1	1									
04-09	CENTER DECK	A240-304L	2T	1	–	–	1				1							
04-09A	SEAL PLATE	A240-304L	2T	4	–	–	4				4							
04-10	TRAY PLATE	A240-321	2T	2	–	–	2							2				
04-11	TRAY PLATE	A240-321	2T	2	–	–	2							2				
04-12	TRAY PLATE	A240-321	2T	2	–	–	2							2				
04-13	TRAY PLATE(M/W)	A240-321	2T	2	–	–	2							2				
04-14	TRAY PLATE	A240-321	2T	2	–	–	2							2				
04-15	TRAY PLATE	A240-321	2T	2	–	–	2							2				
04-16	TRAY PLATE	A240-321	2T	2	–	–	2							2				
04-17	SHIM PLATE	A240-321	2T	2	–	–	2							2				
04-18	LEVELING SHIM	A240-304L	1.5T	112	–	–	112	16	16	16								
04-19	DOWNCOMER BRACKET	A240-304L	3T	28	–	–	28	2	2				2	2				2
04-20	TRAY PLATE	A240-304L	2T	14	–	–	14						2	2				
04-21	SEAL PAN	A240-304L	2T	1	–	–	1								1			
04-22	TRAY PLATE	A240-304L	2T	14	–	–	14						2	2				
04-23	TRAY PLATE	A240-304L	2T	14	–	–	14						2	2				
04-24	TRAY PLATE(M/W)	A240-304L	2T	14	–	–	14						2	2				
04-25	TRAY PLATE	A240-304L	2T	14	–	–	14						2	2				
04-26	TRAY PLATE	A240-304L	2T	14	–	–	14						2	2				
04-27	TRAY PLATE	A240-304L	2T	14	–	–	14						2	2				
04-28	SEAL PAN	A240-304L	2T	1	–	–	1								1			
04-29	SIDE DECK	A240-304L	2T	14	–	–	14						2	2				
04-30	DOWNCOMER BRACE	A240-304L	2T	14	–	–	14						2	2				
04-31	DOWNCOMER BRACKET	A240-304L	3T	2	–	–	2								2			
04-32	TRAY PLATE	A240-321	2T	3	–	–	3									1	1	
04-33	TRAY PLATE	A240-321	2T	3	–	–	3									1	1	
04-34	TRAY PLATE	A240-321	2T	3	–	–	3									1	1	
04-35	TRAY PLATE	A240-321	2T	3	–	–	3									1	1	
04-36	TRAY PLATE(M/W)	A240-321	2T	3	–	–	3									1	1	
04-37	TRAY PLATE	A240-321	2T	3	–	–	3									1	1	
04-38	TRAY PLATE	A240-321	2T	3	–	–	3									1	1	
04-39	TRAY PLATE	A240-321	2T	3	–	–	3									1	1	
04-40	TRAY PLATE	A240-321	2T	3	–	–	3									1	1	
04-42	SHIM PLATE	A240-321	4T	3	–	–	3									1	1	
04-43	CHIMNEY TRAY SECTION	A240-304L	3T	2	–	–	2											2
04-44	CHIMNEY TRAY SECTION	A240-304L	3T	2	–	–	2											2
04-45	INLET WEIR	A240-304L	3T	2	–	–	2											2
04-46	SEAL PLATE	A240-304L	3T	4	–	–	4											4
04-47	RISER SECTION	A240-304L	3T	4	–	–	4											4
04-48	RISER SECTION	A240-304L	3T	4	–	–	4											4
04-49	RISER SECTION	A240-304L	3T	4	–	–	4											4
04-50	OUTLET WEIR	A240-304L	2T	14	–	–	14	2	2	2								
04-51	DOWNCOMER	A240-304L	2T	2	–	–	2	2										
04-52	DOWNCOMER	A240-304L	2T	12	–	–	12	12										
04-53	DOWNCOMER	A240-304L	2T	2	–	–	2	2										
04-54	DOWNCOMER	A240-304L	2T	12	–	–	12		2	2								
04-55	DOWNCOMER	A240-304L	2T	12	–	–	12		2	2								
04-56	OUTLET WEIR	A240-321	2T	2	–	–	2							2				
04-57	DOWNCOMER	A240-321	2T	2	–	–	2							2				
04-58	DOWNCOMER	A240-321	2T	2	–	–	2							2				
04-59	OUTLET WEIR	A240-304L	2T	14	–	–	14						2	2				
04-60	DOWNCOMER	A240-304L	2T	12	–	–	12						2					
04-61	DOWNCOMER	A240-304L	2T	12	–	–	12						2					
04-62	OUTLET WEIR	A240-321	2T	3	–	–	3									1	1	
04-63	DOWNCOMER	A240-321	2T	1	–	–	1									1		
04-64	DOWNCOMER	A240-321	2T	2	–	–	2										1	
04-65	DOWNCOMER	A240-321	2T	2	–	–	2										1	
04-66	DOWNCOMER	A240-304L	2T	2	–	–	2							2				
04-67	DOWNCOMER	A240-304L	2T	2	–	–	2							2				
04-68	FALSE DOWNCOMER	A240-304L	2T	2	–	–	2				2							
04-69	FALSE DOWNCOMER	A240-304L	2T	2	–	–	2				2							
04-70	FALSE DOWNCOMER	A240-321	2T	2	–	–	2							2				
04-71	FALSE DOWNCOMER	A240-321	2T	2	–	–	2							2				

PARTS LIST

PART NO.	DESCRIPTION	MATERIAL	SIZE	NET Q'TY	SPARES (CON. & COMM.)	SPARES (2 YEARS)	TOTAL Q'TY	Q'TY PER INTERNAL

FINAL
2006. 6. 19
HANBAL MASSTECH LIMITED

GENERAL INFORMATION

1. REFERENCE DWG. NO. : VP-6340S-103-C1101-003 ~ 013

REV.	DATE	DESCRIPTION	ORIG.	PREP.	CHKD.	APPD.	AUTH.
F	19/06/06	FINAL	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	
1	23/07/05	CONFIRMED THE COMMENTS	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	
0	13/06/05	FOR APPROVAL	HMT	Y.P.KIM	K.T.MOON	J.H.KWON	

SOUTH PARS GAS FIELD DEVELOPMENT-
PHASES 9 & 10, ASSALUYEH-IRAN
ONSHORE FACILITIES

N.I.O.C
PARS OIL AND GAS COMPANY

PROJECT No.: POGC-760-81-156 | DOC. CLASS. | SCALE : NONE

HMT | OIEC | IOEC

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TITLE:
103-C-101/201
CONDENSATE STABILISER
PART LIST DWG.

DWG No. VP-6340S-103-C1101-013 | SHEET No. 11 OF 11 | REV. F